

CSCE 640 QUANTUM ALGORITHMS · MODULE 12 · LECTURE NOTES

BQP and Its Neighbours

What is known, what is believed, and the difference.

MODULE OUTCOMES

- Define BQP and state its basic properties.
- Place BQP relative to P, NP, and PSPACE.
- Explain why NP-complete problems are not expected to fall.
- Explain why factoring's hardness is unproven.
- State what is known versus believed, accurately.

1 The class

BQP IS WHAT A QUANTUM COMPUTER SOLVES EFFICIENTLY WITH BOUNDED ERROR, AND IT IS REMARKABLY ROBUST

BQP is the class of languages decidable by a uniform family of polynomial-size quantum circuits with error probability at most $1/3$ — which is precisely BPP's definition with quantum circuits substituted (CSCE 658 Module 02, CSCE 637 Module 06).

The error constant does not matter — repetition and a majority vote amplify any constant below $1/2$ to exponentially small, exactly as in the classical case — so $1/3$ is a convention rather than a meaningful choice.

And the gate set does not matter — Solovay-Kitaev converts between any two universal gate sets with polylogarithmic overhead (Module 03 §1) — so the class is independent of the hardware's native operations.

Which makes BQP a robust class rather than a model-dependent artefact — the same robustness that makes P and BPP meaningful objects of study — and it is why the class is worth defining at all, rather than talking about particular machines.

2 Where it sits

Statement	Status
$P \subseteq BPP \subseteq BQP$	Proved, and easy — a quantum circuit can simulate a classical reversible one, and randomness is available.

Statement	Status
$BQP \subseteq PSPACE$	Proved — sum the amplitudes over computational paths one at a time, reusing the space (Module 04 §1's path view, used as a proof technique).
$BQP \neq P$	Unproved. Proving it would imply $P \neq PSPACE$, which is a famous open problem — see the note.
$NP \subseteq BQP$	Believed false , with oracle evidence against and no evidence for (§3).
$BQP \subseteq NP$	Believed false — there is oracle evidence that BQP contains problems outside NP, so the two are believed incomparable.
Factoring is not in P	Unproved, and widely assumed — which is what the whole public-key infrastructure already rests on.

$BQP \neq P$ is unproved, and proving it would separate P from PSPACE — which is a famous open problem, so this one is not falling soon. The BQP-versus-P separation implies $P \neq PSPACE$, by the containment chain above: that is the structural reason nobody expects an unconditional proof that quantum computers are more powerful, and it is worth knowing because it explains why the field's central claim stays conditional.

WHICH MEANS THE SUBJECT'S CENTRAL CLAIM IS CONDITIONAL, AND THAT IS WORTH SAYING PLAINLY

Nobody has proved that a quantum computer is faster than a classical one for any problem — in the non-relativised setting, which is the setting that corresponds to actual computation (Module 05 §4's relativisation gap).

Shor's algorithm beats the best *known* classical factoring algorithm — the number field sieve, subexponential — **and no one has shown that no better classical algorithm exists**, nor is anybody close to showing it.

So the field rests on the same kind of assumption that cryptography does — that a well-studied problem is hard because many people have tried and failed — which is reasonable and is not a proof (CSCE 711 Module 02's framing of exactly this).

And that is entirely normal for complexity theory (CSCE 637 Module 01: almost everything is conditional) — **what is not normal is how often the conditional is dropped** in public accounts of this particular subject, where 'quantum computers are exponentially faster' is stated as fact.

3 NP-complete problems

- **The only general-purpose tool is Grover's algorithm, and Grover is provably quadratic** (Module 06 §4's lower bound) — **so brute-force search over candidate solutions will not do it**, and that part is a theorem rather than a belief.
- **And NP-complete problems have no known structure of the kind the QFT exploits — no periodicity, no hidden abelian subgroup** (Module 07 §3) — which removes the only other route to an exponential advantage that anybody has found.
- **Plus there is oracle evidence against: relative to a random oracle, NP is not contained in BQP — which is suggestive rather than conclusive**, for the relativisation reasons of Module 05 §4, but it is evidence pointing one way and none pointing the other.
- **So the belief is well-founded and unproved — and the flat statement 'quantum computers cannot solve NP-complete problems' overstates what is known, which this course does not do** even though the overstatement is on the side of caution.
- **The correct statement is: no approach is known, and the known approaches provably do not suffice — which is the precise form of the claim and is weaker than the usual paraphrase**, and is also more informative, because it says where a surprise would have to come from.

4 Reading the literature

DISTINGUISH PROVED, BELIEVED WITH EVIDENCE, AND ASSUMED FOR CONVENIENCE

Proved: Grover's optimality; the query separations of Module 05; $BQP \subseteq PSPACE$; the threshold theorem — these are theorems and will not change, and they are the firm ground to reason from.

Believed with evidence: NP is not contained in BQP; factoring is classically hard; BQP is strictly larger than P — supported by oracle results and by sustained failure to find counterexamples, which is real evidence of a kind, and is not proof.

Assumed for convenience: that a particular classical baseline is the best available one — which is the assumption that has failed most often in practice (Module 13 §3's pattern of advantage claims being reduced by better classical algorithms), and is the one most under the author's own control.

Which is the program's closing rule applied to a literature: state what you proved, state what you assumed, and keep the conditional attached when you report the result — because the conditional is where the honest content is.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Bernstein & Vazirani — Quantum complexity theory	§1 — where BQP is defined and its robustness established.
Aaronson — BQP and the Polynomial Hierarchy (free)	§2's containments — and the oracle evidence for BQP containing things outside NP.
The Complexity Zoo (free)	§2 — look up BQP and follow the containments, which is the fastest way to see the landscape.
Aaronson — The Limits of Quantum Computers (free)	§3 — the NP question treated carefully and for a general audience, which is harder than it sounds.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Define BQP** and show the error constant is arbitrary.
- 2 **Explain why the gate set does not matter.**
- 3 **Prove $P \subseteq BQP$** , informally.
- 4 **Sketch the $BQP \subseteq PSPACE$ argument** as a path sum.
- 5 **Explain why $BQP \neq P$ would separate P from PSPACE.**
- 6 **State what Shor's algorithm actually establishes**, with the conditional.
- 7 **Give three reasons NP-complete problems are not expected to fall.**
- 8 **State the claim in its precise form**, and contrast with the usual paraphrase.
- 9 **Classify ten claims** from this course as proved, believed, or assumed.
- 10 **Find a paper that drops a conditional** in its abstract but keeps it in the body.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Define BQP and give its two robustness properties.
- 2 Which containments are proved?
- 3 Why is $BQP \neq P$ hard to prove?
- 4 What has nobody proved about quantum speedups?
- 5 What does Shor's algorithm establish, exactly?
- 6 What kind of assumption does the field rest on?

- 7 Give three reasons NP-complete problems are not expected to fall.
- 8 State the precise claim about NP-complete problems.
- 9 Give the three epistemic categories with an example of each.
- 10 Which category has failed most often in practice?